David Ramirez Betancourth

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Professional Profile

Electronic Engineer specialized in Radio Frequency (RF) systems and Digital Signal Processing (DSP). Experience in hardware-software integration from low-level protocols (TCP, UDP, ZMQ) and Software Defined Radio (SDR) to cloud architectures with IoT sensors. Firmware development in C for real-time acquisition and processing, with autonomous calibration and Python orchestration. I complement my development workflow with AI agents for TDD/SDD automation, focusing on high-performance embedded architecture optimization.

Education

MSc **National University of Colombia**, Early admission to M.Sc. in Industrial Automation — Jan 2026 - Present

- **Research Areas:** Artificial Intelligence, Machine Learning applied to RF, Computational Structures, and Optimization.

BS **National University of Colombia**, Electronic Engineering — Jul 2022 - Jun 2026

- **Cumulative GPA:** 4.2/5.0
- **Focus Areas:** Digital Signal Processing (DSP), Software Defined Radio (SDR), Hardware Design, and IoT Systems.

Experience

GCPDS Research Group, Researcher and Developer — Manizales, Colombia, Oct 2024 – Present

- C/Python firmware development for an IoT spectral monitoring sensor based on HackRF SDR on Raspberry Pi 5, with hybrid architecture (C for real-time IQ acquisition, Python for orchestration), autonomous PPM calibration, and WebRTC streaming.
- Backend platform implementation with REST API (FastAPI), TimescaleDB time-series database, and real-time spectral visualization dashboard.
- Mobile application development for deployment and validation of Image Segmentation models.

National University of Colombia, Undergraduate Fellow — Manizales, Colombia, 2024

- Academic teaching assistant for Signals and Systems and Machine Learning Theory, advising groups of 30+ students in theoretical and practical components.
- Design and instruction of hands-on Python workshops for signal analysis and processing, integrating tools such as NumPy, SciPy, and Matplotlib.

Projects

IoT Spectrum Monitoring Sensor — Repo: SDR-Sensor

- Firmware development for a spectral monitoring system integrating Raspberry Pi 5 and HackRF SDR. Includes implementation of calibration protocols, precise NTP synchronization, and optimized drivers for SDR hardware.
- **Tools:** C, Python, HackRF, Raspberry Pi.

RPI-DashWake (Smart Clock) — Repo: RPI-DashWake

- Design of a smart clock with integrated environmental monitoring (temperature,

humidity, and air quality). Includes a web server for real-time metric visualization and programmable alarm management.
- **Technologies:** C, Python, Arduino, JavaScript.

SelfHostedContextCrafter (AI Agent Orchestrator) Repo: ContextCrafter

- Automated architectural team that orchestrates RAG-based expert agents, running fully on local hardware. It digests hardware datasheets, software documentation, and network protocols to generate structured context and automate development workflows.
- **Technologies:** Python, RAG, local LLMs, LaTeX.

Technical Achievements

- **Agnostic SDR architecture:** Architecture design for a cognitive sensor network, integrating WebRTC for real-time streaming, TimescaleDB for historical data, and high-performance WebGL visualization.
- **DSP and SDR calibration:** Development of iterative calibration algorithms (gradient descent) to correct clock offset (PPM) in SDR hardware, and implementation of custom DSP engines in C99 using Polyphase Filter Banks (PFB) and *liquid-dsp*.
- **Telemetry and profiling on Pi 5:** Execution of hardware stress tests and power profiling (PD/PPS configurations) on Raspberry Pi 5, identifying and mitigating voltage drops under concurrent LTE and radiofrequency processing workloads.
- **AI automation (TDD/SDD):** Development workflow automation using AI agents with TDD and SDD methodologies, generating code, tests, and documentation predictively. Significant reduction in repetitive boilerplate and validation tasks.
- **Agent orchestrator (SelfHostedContextCrafter):** Development of a system that orchestrates RAG-based expert agents on local hardware, capable of digesting hardware datasheets and technical documentation to generate structured context and automate code review and deployment tasks.

Technologies

Languages: C99, C++, Python, JavaScript, TypeScript, Verilog, SQL

Frameworks and Platforms: FastAPI, Django, React, Vite, Node.js, GNU Radio, SoapySDR, HackRF One, FPGAs (Gowin EDA)

Infrastructure and Tools: Linux (systemd, Bash), Docker, CMake, ZMQ, WebRTC, TimescaleDB, Git, Sphinx, Doxygen, NTP, Raspberry Pi, GPS/LTE

Languages

Spanish: Native | **English:** Intermediate (B1)